Spherical Splits for Decision Trees and Random Forests: A Short Note

Abstract

Classical decision trees split covariate space with coordinate thresholds, which creates rectangular recursive partitions. This note studies a simple alternative in which a node sends observations left or right according to whether they lie inside a learned hypersphere. The construction preserves the usual impurity-based tree architecture while replacing some threshold questions by distance-to-center questions. We describe the split rule for a data matrix and response vector, show that distant sphere centers recover linear frontiers locally, and note that CART cost-complexity pruning applies without changing the pruning criterion. Exploratory benchmarks on PMLB datasets suggest that spherical random forests can improve mean classification performance, although the current split search is slower and the same advantage is not observed for regression or for single trees.

Keywords: decision trees; random forests; classification; spherical splits; recursive partitioning.

1 Introduction

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a data matrix, with rows $\mathbf{x}_i^\top$, and let $\mathbf{y}$ be a class label or response vector. Decision trees remain useful for such tabular classification and regression problems because they are nonparametric, interpretable, and easy to ensemble. Their usual form, however, fixes a strong geometry. At each internal node, CART-style trees, as introduced by Breiman, Friedman, Olshen, and Stone (1984), select a feature index j and a threshold a, then split observations according to $x_{ij} \leq a$ or $x_{ij} > a$. Recursively, this creates rectangular cells.

Rectangular cells are natural when the signal is sparse and coordinate-aligned. They are less economical when the target depends on membership in an area around a center. In classification, a class may correspond to a compact region or cluster; in regression, a response surface may change near a centroid or prototype. An axis-aligned tree can approximate such a region, but it may need many cuts. A spherical split can isolate the same type of region directly.

The resulting rule can also be easier to read than an oblique split in some settings. An oblique rule asks whether a weighted linear combination of covariates exceeds a threshold, as in the OC1 system of Murthy, Kasif, and Salzberg (1994). A spherical rule asks whether an observation lies within radius r of a center $\mathbf{c}$. When the active covariates define a meaningful metric space, this is a local-neighborhood statement. For example, with projected latitude and longitude, a split describes membership in a circular region around a geographical point. This interpretation depends on scaling and on the metric, but it can be more direct than an abstract linear score.

This short note makes three points. First, spherical splits define a simple finite impurity search that can be inserted into ordinary tree induction. Second, they interpolate geometrically between radial rules and linear frontiers. Third, post-hoc

CART pruning applies without changing the pruning criterion. The empirical results are presented as exploratory evidence, not as a dominance claim.

2 Spherical split rule

At a node, let $S \subseteq \{1, \dots, p\}$ be the active feature set. A spherical split is determined by a center $\mathbf{c} \in \mathbb{R}^{|S|}$ and a radius $r \geq 0$:

$$\|\mathbf{x}_{iS} - \mathbf{c}\|_2^2 \leq r^2.$$

Observations satisfying the inequality are sent to one child and the remaining observations to the other. The pair $(\mathbf{c}, r)$ is chosen to maximize the same impurity decrease used in ordinary tree induction, such as Gini decrease for classification or variance reduction for regression.

The implementation studied here uses finite candidate sets. Candidate centers are generated from global centers, target-conditional centers, observed points, pairwise midpoints, local Gaussian perturbations, and farther radial perturbations. For each candidate center, the candidate radii are the Euclidean distances from the center to the node samples. Thus the geometry changes, but the optimization remains a finite sequence of impurity comparisons.

Figure 1 compares axis-aligned, oblique, and spherical partitions on two-dimensional examples. The spherical panels show internal circles; terminal regions are intersections of recursive inside/outside decisions.

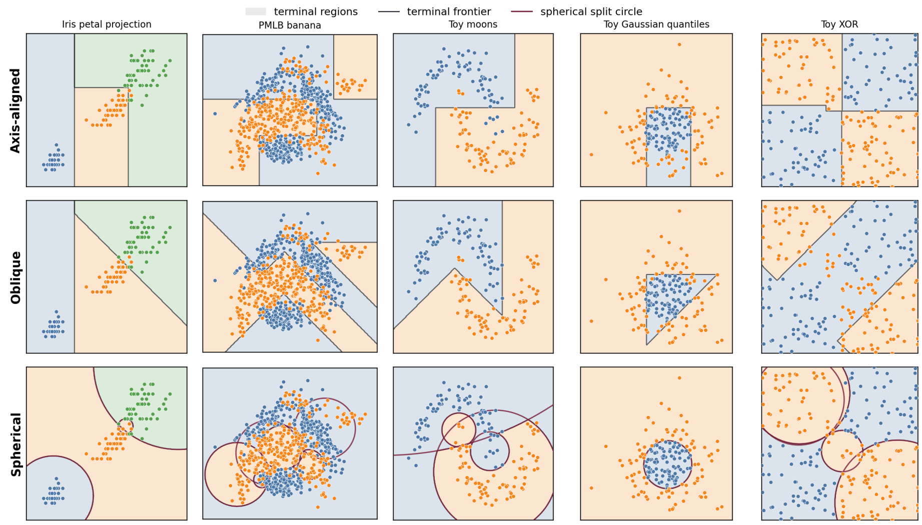

Figure 1: Two-dimensional decision regions learned by shallow axis-aligned, oblique, and spherical trees. Black curves are terminal decision frontiers; burgundy circles are internal spherical split boundaries.

3 Geometry and pruning

The most direct case for spherical splits is radial or local structure. If the target changes when $\mathbf{x}$ enters a ball around $\mathbf{c}$, then one spherical split can represent that event. A rectangular tree must approximate the same ball by coordinate cuts.

Spherical splits also recover linear frontiers as a local limit. Let $\mathbf{u} \in \mathbb{R}^p$ satisfy $\|\mathbf{u}\|_2 = 1$, fix $t \in \mathbb{R}$, let R be large enough that $R^2 - 2Rt \geq 0$, set $\mathbf{c} = R\mathbf{u}$, and choose

$$r_R^2 = R^2 - 2Rt.$$

Then

$$\begin{aligned}\|\mathbf{x} - R\mathbf{u}\|_2^2 \leq r_R^2 &\iff \|\mathbf{x}\|_2^2 - 2R\mathbf{u}^\top\mathbf{x} + R^2 \leq R^2 - 2Rt \\ &\iff \mathbf{u}^\top\mathbf{x} \geq t + \frac{\|\mathbf{x}\|_2^2}{2R}.\end{aligned}$$

On any bounded region $\|\mathbf{x}\|_2 \leq B$, the final term is at most $B^2/(2R)$. Hence the spherical split converges locally to the halfspace $\mathbf{u}^\top\mathbf{x} \geq t$ as the center is moved far away. Taking $\mathbf{u}$ to be a coordinate vector recovers an axis-aligned split in the same limiting sense.

A leaf at depth d is an intersection of at most d sets, each either a ball or the complement of a ball. Spherical-tree leaves are therefore semi-algebraic cells defined by quadratic inequalities. In two dimensions their boundaries are circular arcs; in higher dimensions they are hyperspherical patches. This differs from axis-aligned boxes and oblique polyhedra, and it can represent annuli or shells compactly.

Cost-complexity pruning adapts directly. CART pruning selects subtrees by minimizing

$$R_\alpha(T) = R(T) + \alpha|\widetilde{T}|,$$

where $R(T)$ is the empirical leaf risk and $|\widetilde{T}|$ is the number of terminal nodes (Breiman et al., 1984). Once a spherical tree has been grown, this criterion depends only on topology, leaf predictions, leaf risks, and leaf count. It does not depend on the internal split geometry. The usual weakest-link pruning sequence and validation choice of α can therefore be used unchanged.

4 Exploratory benchmark

We implemented spherical trees and spherical random forests in a `treeple`-based prototype. We compared them with axis-aligned CART and random forests, as well as oblique trees and forests. The benchmark used complete PMLB datasets of Olson, La Cava, Orzechowski, Urbanowicz, and Moore (2017) that ran without errors. Features were imputed and standardized. The comparison used three cross-validation folds, 50 trees for forests, 500 spherical center candidates per node, all data-induced radii for each candidate center, validation-selected pruning for single trees, and un-

Table 1: Mean performance and fit time over complete datasets. The primary score is balanced accuracy for classification and R^2 for regression.

Task	Model	Score	Fit time (s)
Classification	Spherical random forest	0.765	9.850
Classification	Random forest	0.738	0.137
Classification	Oblique random forest	0.737	0.154
Classification	CART	0.722	0.074
Classification	Oblique tree	0.706	0.136
Classification	Spherical tree	0.688	4.132
Regression	Random forest	0.717	0.384
Regression	Oblique random forest	0.700	0.434
Regression	Spherical random forest	0.665	20.533
Regression	CART	0.521	0.078
Regression	Oblique tree	0.391	0.093
Regression	Spherical tree	0.262	7.516

pruned forests. Large datasets were capped at 1000 sampled observations. The final comparison contained 249 complete datasets and 38 skipped datasets or fetch/model errors.

The salient empirical pattern is the classification performance of spherical forests. In this exploratory benchmark, spherical random forests obtained the best mean classification score among the compared forest methods. The advantage does not appear generally in regression, where ordinary and oblique forests remain stronger. Nor does it appear on average for single trees. This suggests a specific use case: high-variance spherical split candidates may be useful when stabilized by ensembling, especially in classification problems with local or cluster-like structure.

Figure 2 also shows the practical caveat. At a node with n_m samples, s active features, and C candidate centers, the basic distance computation costs $O(Cn_m s)$. If all data-induced radii are considered, sorting distances for each center adds work of order

$$O(Cn_m \log n_m),$$

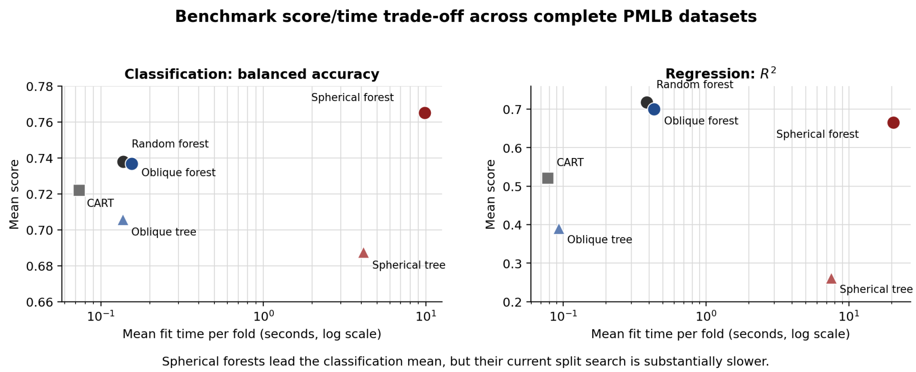

Figure 2: Mean score and mean fit time over complete PMLB datasets. The current spherical split search improves the classification mean for forests but is substantially slower; in regression, ordinary and oblique forests remain stronger and faster.

after which cumulative impurity updates scan the ordered candidates. Faster matrix-based distance evaluation, approximate radius grids, and node-level parallelism are natural implementation targets.

5 Discussion

Spherical splits should not be read as a replacement for CART. They are a complementary local geometry. They are most appealing when the target is expected to depend on membership in an area of covariate space, especially around a center, prototype, cluster, or spatial location. They also connect to linear splitting through distant centers, but this does not remove the greedy search limitation of tree induction. In the XOR example, for instance, the two axis-aligned cuts are individually weak at the root even though they are useful together; a depth-two lookahead would be needed to systematically recover such interactions.

The method also depends on the metric. In our experiments, predictors were standardized before fitting. Without scaling, a spherical split is dominated by high-variance coordinates. In high-dimensional spaces, Euclidean distances can become less informative, and far-away centers may make splits nearly linear without solving the underlying greedy-search problem.

The main conclusion is modest: spherical splits provide an interpretable and geometrically distinct split family for recursive partitioning. They inherit CART pruning, include linear frontiers as a local limiting case, and show promising exploratory behavior for random-forest classification. The next methodological step is a hybrid tree that chooses among axis-aligned, oblique, and spherical candidates at each node, possibly with a small lookahead window.

Data availability

The benchmark reuses public PMLB datasets (Olson et al., 2017). Dataset names, skipped-dataset logs, and the stored benchmark summaries are included in the supplementary reproduction materials.

Code availability

Custom code and reproduction scripts are supplied as supplementary material for peer review. The supplementary materials include `REPRODUCING_RESULTS_BLINDED.md`, which documents how to retrieve stored result tables and regenerate all figures and results reported in this short note. Upon publication, the code will be made available in a public repository.

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